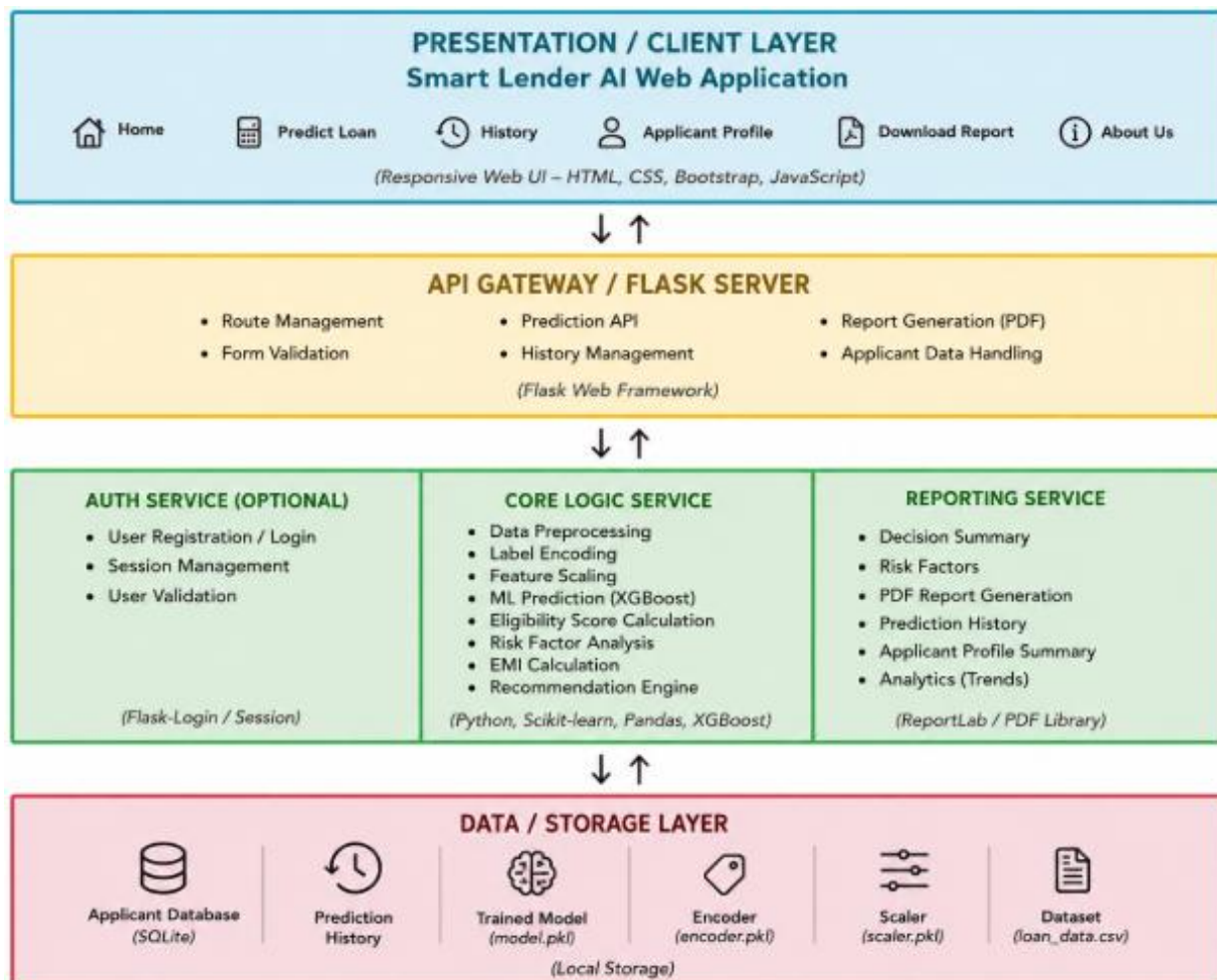


Solution Architecture

Date	10 Jul 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation / Client Layer</i>	Provides the user interface for loan prediction, applicant profile, history, and reports.	<i>HTML, CSS, JavaScript, Bootstrap</i>
<i>Application / API Layer</i>	<i>Handles user requests, validates input, processes predictions, and manages application flow.</i>	Flask, Python
<i>Core Logic Service</i>	<i>Performs data preprocessing, feature scaling, machine learning prediction, eligibility score calculation, and recommendation generation.</i>	Scikit-learn, XGBoost, Pandas, NumPy
<i>Data / Storage Layer</i>	Stores applicant data, prediction history, trained models, encoders, scalers, and loan dataset.	<i>SQLite, Joblib (.pkl), CSV Files</i>